KOFORIDUA TECHNICAL UNIVERSITY

FACULTY OF HEALTH AND ALLIED SCIENCES

DEPARTMENT OF BIOMEDICAL ENGINEERING TECHNOLOGY

TOPIC

EXAMINING THE TECHNICAL CHALLENGES BIOMEDICAL ENGINEERING STUDENTS ENCOUNTER DURING CLINICAL TRAINING AND THEIR IMPACT ON PROFESSIONAL DEVELOPMENT IN GHANA

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STUDENT'S DECLARATION

I hereby declare that this research is the result of my own original research and that no part of it has been presented for another certificate in this Institution or elsewhere.

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SUPERVISOR'S CERTIFICATION

I hereby certify that this project work was supervised in accordance with the University's guidelines for supervision of project work.

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ABSTRACT

Biomedical Engineering (BME) plays a vital role in strengthening Ghana's healthcare system by ensuring that medical technologies are properly designed, maintained, and applied in patient care. For students training in this field, clinical rotations are meant to bridge theory and practice by providing hands-on experience with real hospital equipment. However, many students struggle to gain this experience because of challenges such as limited access to functional equipment, inadequate supervision, and restricted opportunities for independent practice. This study set out to explore the technical challenges faced by biomedical engineering students during clinical training and to understand how these challenges affect their professional growth. A descriptive cross-sectional survey was carried out among 120 students across four universities, using structured questionnaires and descriptive analysis. The results showed that 55% of students reported equipment was not always available, while 48% noted that some devices were non-functional when needed. Technical difficulties were also common: 62% of students struggled with operating clinical devices, 58% experienced poor supervision, and 65% lacked adequate hands-on practice. These difficulties had serious consequences for professional development, with 60% of students reporting reduced confidence, 57% pointing to gaps in competence, and 54% feeling unprepared for real-world practice. The findings underscore the need for hospitals and universities to work together by providing reliable equipment, preventive maintenance, and structured mentorship. Addressing these gaps will not only enhance students' skills and confidence but also ensure Ghana produces biomedical engineers who are ready to contribute effectively to healthcare delivery.

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LIST OF ABBREVIATIONS

Accra Technical University
(ATU) 11
All Nations University
(ANU)..... 11
Biomedical Engineering
(BME)..... 11
Community based Health Planning and Services
(CHPS) 18
Ghana Tertiary Education Commission
(GTEC)..... 13
Koforidua Technical University
(KTU) 11
Kwame Nkrumah University of Science and Technology
(KNUST)..... 11
Ministry of Health
(MOH)..... 19
University of Ghana
(UG) 11
World Health Organization
[WHO]..... 12

CHAPTER ONE

10. INTRODUCTION

1.1 Background of the Study

Biomedical Engineering (BME) is a rapidly growing discipline that merges principles of engineering with medical sciences to design, develop and maintain healthcare technologies. In many parts of the world, BME professionals play an integral role in ensuring that medical equipment functions optimally to support diagnosis, treatment, and patient care (Enderle, 2022). Biomedical engineering students, therefore, require a training model that blends theoretical knowledge with practical clinical experiences to prepare them for professional practice.

In Ghana, Biomedical Engineering as an academic discipline began to emerge in the early 2000s, when Kwame Nkrumah University of Science and Technology (KNUST) pioneered the first undergraduate program. Over the years, other institutions such as the University of Ghana (UG), Accra Technical University (ATU), Koforidua Technical University (KTU), and All Nations University (ANU) have also established accredited BME programs. The introduction and expansion of these programs were intended to bridge the growing gap between the healthcare system's increasing reliance on technology and the workforce capable of managing and sustaining it (Amankwah & Frimpong, 2020). This growth reflects both national and global recognition of the vital role biomedical engineers play in supporting healthcare delivery.

Clinical training forms a crucial part of this education, as it provides students with exposure to hospital environments, hands-on interaction with medical equipment, and opportunities to develop the confidence and competencies required for independent practice. In low and middle-income countries like Ghana, students often face technical and systemic challenges during clinical rotation. However, the effectiveness of clinical training in Ghana is sometimes constrained by resource limitations. Medical equipment in hospitals and training institutions is often either unavailable, obsolete, or non-functional due to poor maintenance and inadequate budgets (Boateng *et al*., 2019). Consequently, biomedical engineering students are unable to fully practice their skills in real-world settings. Limited access to modern diagnostic and therapeutic equipment such as patient

monitors, ECG machines, infusion pumps, suction machines and sterilisation units hinders their ability to troubleshoot, maintain, and calibrate devices effectively.

According to Kolb's theory of experiential learning cycle, learning is most effective when learners can reflect on and apply hands-on experience (Beard & Wilson, 2018). Technical challenges during clinical training not only affect immediate learning outcomes but can also hinder students' transition into competent biomedical engineering professionals. This gap impacts healthcare systems, especially in Ghana, where biomedical engineers play a crucial role in maintaining and innovating health technologies (WHO, 2017).

Globally, the literature emphasises that technical challenges during training affect professional identity formation, confidence, and readiness for practice (World Health Organisation [WHO], 2021). For biomedical engineers, this translates into delayed acquisition of critical skills, diminished confidence in handling medical devices, and limited preparedness for addressing healthcare technology challenges after graduation. This concern is particularly pressing in Ghana, where the healthcare system already faces significant challenges related to equipment shortages, frequent breakdowns, and a lack of technical expertise (Darko, 2021).

Thus, studying the technical challenges biomedical engineering students encounter during clinical training and how these challenges affect their professional development is vital. Such research provides evidence to improve training curricula, strengthen clinical partnerships between universities and hospitals, and contribute to producing competent biomedical engineers who can support Ghana's healthcare needs.

1.2 Problem Statement

Despite the growing needs of Ghana's healthcare system for competent biomedical engineers, students in clinical training face persistent challenges such as limited access to functional equipment, inadequate supervision and restricted hands-on practice. These difficulties hinder their acquisition of critical skills in troubleshooting, calibration and maintenance, ultimately reducing their confidence and readiness for professional roles. Yet, there is limited empirical data that systematically documents these challenges and their impact on students' professional development. This study seeks to fill that gap by examining the technical challenges biomedical

engineering students encounter during clinical training and how these affect their competence and confidence.

1.3 Research Aim and Objectives

1.3.1. Aim

The main aim of this study is to examine the technical challenges biomedical engineering students encounter during clinical training and their impact on professional development in Ghana.

1.3.2 The specific objectives:

a) To assess the availability and functionality of medical equipment used during students' clinical training.
b) To identify the technical challenges biomedical engineering students encounter during clinical training in healthcare settings.
c) To analyse how these challenges affect students' skills acquisition, confidence and readiness for professional practice.

1.3.3 Research Questions

a) Which equipment were available and functioning?
b) What technical challenges did they students encounter?
c) How did these challenges affect their readiness for professional practice?

1.4 Justification of the Study

Academic Contribution:

The study contributes to the limited body of literature on biomedical engineering education in Ghana. While many studies have examined medical education and nursing training, very few focus on biomedical engineering students and their clinical experiences. By addressing this gap, the research enriches academic discussions on engineering-in-healthcare education in Africa (Abubakari, 2019).

Policy Relevance:

Findings will provide evidence-based recommendations for the Ministry of Health, Ghana Tertiary Education Commission (GTEC), and university faculties on how to strengthen biomedical engineering training. Policymakers can use the results to improve resource allocation for medical equipment in teaching hospitals and revise curricula to better prepare students for real-world challenges.

Practical Relevance:

By highlighting specific technical challenges (e.g., lack of functional equipment, poor supervision, limited hands-on training), the study can help hospitals and training institutions develop targeted interventions. This may include preventive maintenance strategies, structured mentorship and better collaboration between universities and healthcare facilities.

Student Development:

Biomedical engineering students stand to benefit directly from the study. Identifying the gaps in their training environment will help universities create more supportive learning conditions, boosting students' confidence, skills, and professional identity.

National Development:

Competent biomedical engineers are vital to Ghana's broader health agenda, including the attainment of Sustainable Development Goal 3 (Good Health and Well-being). By improving the training of biomedical engineers, Ghana strengthens its healthcare system's resilience, reduces equipment downtime, and ensures better patient outcomes.

CHAPTER TWO

2.0 LITERATURE REVIEW

2.1 Introduction

The study starts with comprehensive review of published works on clinical rotation in healthcare facilities. The main sources of literature include International Journal of Biomedical Engineering, published thesis and data from students and hospitals. This provides vital inputs for the research design and implementation of the specific research activities.

2.2. Biomedical Engineering Education

2.2.1 Global Perspective on Biomedical Engineering Education

Biomedical Engineering education has become a cornerstone of healthcare technology innovation worldwide. In countries such as the United States, Germany and Japan, BME programs are well established, offering comprehensive curricula that combine engineering principles with clinical applications (Enderle & Bronzino, 2020). Institutions such as Johns Hopkins University and MIT have pioneered models of education that integrate research, clinical immersion and industry collaboration to equip students with practical and problem-solving skills (Linsenmeier, 2003).

A hallmark of global BME education is the incorporation of clinical training and hands-on experience through internships and capstone design projects. These experiences are designed to bridge the gap between theoretical knowledge and practical application (Gassert & Enderle, 2009). In Europe for example, BME education is aligned with the Bologna Process, ensuring consistency in academic standards and mobility across institutions (Iadanza, 2018).

Despite the progress, challenges persist globally. These include the rapid pace of technological change which often outstrips curriculum updates and the need for multidisciplinary collaboration which can be difficult to implement effectively in rigid academic structures (Harris *et al.*, 2013).

2.2.2 Biomedical Engineering Education in West Africa

Biomedical Engineering is still in a developmental phase in many West African countries. Most universities offering BME programs are under-resourced with limited faculty expertise, outdated laboratory equipment and few partnerships with the healthcare sector (Adeleke *et al.*, 2020). BME curricula often emphasize theoretical instruction with minimal emphasis on practical or clinical training.

In Nigeria for example, institutions such as the University of Lagos and Obafemi Awolowo University have launched BME programs but practical training is hindered by poor access to clinical environments and malfunctioning medical equipment (Adeyemi & Obafemi, 2022). Clinical rotations where they exist, tend to be informal and lack structured supervision.

A major issue in West African BME education is the absence of a harmonized framework for training and certification. Unlike Europe or North America, West African nations do not have regional accreditation bodies for BME education, leading to disparities in program quality and graduate competence (Oluwafemi & Adebayo, 2019). Furthermore, collaboration between universities and healthcare facilities is often weak, which limits students' exposure to hospital technologies and maintenance practices (Ajayi *et al*., 2021).

2.2.3 Biomedical Engineering Education in Ghana

In Ghana, Biomedical Engineering is a growing academic and professional field. Institutions like the Kwame Nkrumah University of Science and Technology (KNUST) and the University of Ghana have established undergraduate and graduate BME programs to address local healthcare technology needs (Nyamekye & Darko, 2020). These programs are designed to produce graduates capable of managing, repairing and innovating medical technologies in the country's healthcare system.

However, several challenges affect the effectiveness of BME education in Ghana. First, there is limited access to functioning biomedical equipment in both university laboratories and clinical training centers. Many hospitals lack the infrastructure or willingness to host engineering students for hands-on training, citing liability issues and equipment fragility (Asare & Baffour-Awuah, 2021).

Second, clinical rotations are often not formalized and when they occur, they are poorly supervised. Students frequently report limited guidance from clinical engineers or hospital staff which affects their ability to gain practical skills (Tetteh *et al*., 2023). Third, the alignment between the academic curriculum and the realities of the job market is weak. Many students graduate without the practical competencies needed for employment in hospitals, leading to skills mismatches (Owusu-Antwi & Amponsah, 2022).

Despite these issues, stakeholders in Ghana have begun recognizing the need for curriculum reform and stronger industry-academic partnerships. The Ministry of Health and Ghana Health Service have shown interest in collaborating with academic institutions to improve clinical training infrastructure and standardize internship programs for BME students (World Bank, 2019).

2.3 Scope of Clinical Rotations and Internships in Healthcare Facilities

Biomedical Engineering Students' Clinical Rotations or Internships in Healthcare Facilities. Clinical rotations or internships form a vital part of biomedical engineering (BME) education, serving as a bridge between theoretical knowledge and practical application in real-world healthcare settings. These internships are typically coordinated by academic institutions in partnership with hospitals, diagnostic centers and medical equipment manufacturers to provide students with hands-on experience in equipment management, device calibration, installation, troubleshooting and maintenance (Ogundiran *et al*., 2020; Appiah-Kubi *et al*., 2023).

In Ghana, clinical rotations for BME students are usually organized during the penultimate or final year of study. Academic institutions such as Koforidua Technical University (KTU), All Nations University, Kwame Nkrumah University of Science and Technology (KNUST), University of Ghana, Valley View University, Academic City University College and Accra Technical University have established internship programs in collaboration with hospitals like Korle Bu Teaching Hospital, Greater Accra Regional Hospital and Komfo Anokye Teaching Hospital. However, the structure, supervision and quality of these internships vary significantly depending on the institution and host facility (Bediako & Sarpong, 2020).

According to Adusei et al. (2021), a key challenge is the lack of standardized internship frameworks or national guidelines that define learning objectives, mentorship responsibilities and performance evaluation criteria for BME students. This often results in students being

underutilized or assigned administrative rather than technical roles. In some cases, host institutions are not fully aware of the academic goals of the rotation, limiting the student's ability to engage with high-risk or advanced medical equipment.

Moreover, a study by Mensah and Osei (2020) found that while students appreciate the opportunity to witness clinical procedures and interact with hospital engineers, many report feeling disconnected from the actual biomedical functions due to limited mentorship or poor integration into the hospital's technical team. The same study emphasized the importance of continuous feedback, formal orientation sessions and exposure to a wide range of medical technologies to enhance internship effectiveness.

The quality of the internship experience also has long-term implications for professional readiness. Clinical rotations not only expose students to the technical realities of the job but also help them develop soft skills such as communication, teamwork and adaptability, which are essential in healthcare settings (Nzambi & Tetteh, 2023). Thus, ensuring that BME internships are well-structured, adequately supervised and aligned with educational objectives is key to building a competent and confident biomedical engineering workforce.

2.4 Level of Healthcare Systems and Types Medical Equipment Used in Ghana

2.4.1 Level of Healthcare Systems

Ghana's healthcare system is structured into three main levels that is the primary, secondary, and tertiary each serving a unique role in healthcare delivery. The primary level consists of Community based Health Planning and Services (CHPS) compounds, health centres and polyclinics. These facilities mainly provide preventive and basic curative services, often equipped with only basic diagnostic and therapeutic tools like stethoscopes, sphygmomanometers and basic laboratory equipment (Ghana Health Service [GHS], 2022).

The secondary level includes district and regional hospitals, which offer more specialised diagnostic and therapeutic care. These facilities typically possess a broader range of medical equipment such as X-ray machines, ultrasound scanners, ECG machines and basic surgical tools (World Health Organisation, WHO, 2020). However, operational inefficiencies, poor maintenance and equipment shortages remain challenges, especially in rural regions (Boateng *et al.,* 2021).

At the tertiary level, Ghana's teaching hospitals such as Korle Bu, Komfo Anokye and Tamale Teaching Hospital provide advanced care and serve as training centres for healthcare professionals, including biomedical engineering students. These institutions are often equipped with high-end medical devices like MRI and CT scanners, ventilators, infusion pumps and sophisticated diagnostic equipment (Ministry of Health, MOH, 2021). Despite this, even tertiary facilities face periodic breakdowns and resource constraints that impact the learning experience of students during clinical rotations (Appiah, 2020).

2.4.2 Types of Medical Equipment Used in Ghana

The availability and type of medical equipment vary across healthcare levels and directly affect the technical exposure of students. Diagnostic equipment includes devices such as ECG machines, X-ray machines, and ultrasound systems, commonly found in secondary and tertiary facilities. However, due to limited maintenance and calibration, their functionality can be compromised (Owusu *et al.,* 2022).

Therapeutic and life-support equipment, such as defibrillators, suction machines, infusion pumps and ventilators, are essential for critical care but are not evenly distributed across facilities. Rural and under-resourced hospitals often lack such tools or use outdated models (WHO, 2020). Monitoring equipment like patient monitors and pulse oximeters are prevalent in emergency and surgical units, yet students report limited access due to priority use by staff.

In terms of surgical and sterilisation equipment, tertiary and some regional hospitals possess autoclaves, anaesthesia machines and surgical tables but functionality and availability are inconsistent. Biomedical engineering students often face broken or decommissioned machines reducing their opportunity for hands-on technical practice (Boateng *et al*., 2021).

2.5 Medical Equipment Uptime

The availability and operational status of medical equipment play a pivotal role in the clinical training of Biomedical Engineering (BME) students. Clinical exposure provides students with the hands-on experience necessary to understand, maintain and troubleshoot medical technologies (Ogundiran *et al.,* 2020). However, in many developing countries, including Ghana, there is often limited access to modern, fully functional equipment in public hospitals due to procurement delays, funding constraints and poor maintenance culture (Adusei *et al.,* 2021).

A study by Boateng and Badu (2022) in selected Ghanaian teaching hospitals revealed that over 35% of essential biomedical equipment were either obsolete or non-functional during clinical placements. This limitation not only hampers the training experience but also reduces students' ability to engage with a wide spectrum of technologies, especially high-end diagnostic and therapeutic devices such as CT scanners and radiotherapy systems.

Moreover, healthcare institutions in Ghana often lack dedicated biomedical engineering departments or structured internship protocols, which means students may not always be paired with experienced mentors or provided with equipment manuals and learning modules (Asare *et al.,* 2023). The inadequacy in infrastructure further discourages practical learning and restricts students to passive observation.

2.6 Technical Challenges in Clinical Training

Biomedical engineering students face several technical challenges during clinical training. Among the most common are limited access to real-time fault diagnostics, absence of structured technical supervision and poor integration between hospital staff and interns (Mensah & Osei, 2020). Students often report inadequate orientation to hospital equipment policies, leaving them unsure of their boundaries and roles (Ampofo & Nketiah, 2021).

Furthermore, issues such as lack of spare parts, frequent power outages and incompatibility of donated equipment with local systems contribute to technical frustrations (World Health Organisation [WHO], 2022). These conditions make it difficult for students to develop technical problem-solving skills. Inadequate documentation on past equipment faults and lack of exposure to standardised troubleshooting protocols were also cited as barriers by Dapaah *et al*. (2021) in a study involving BME interns from three universities in Ghana.

Gender dynamics may also play a role, with female students facing additional challenges such as underrepresentation in technical discussions and task assignments, as reported in recent studies on engineering training programs in sub-Saharan Africa (Nzambi & Tetteh, 2023).

2.7 Skills Acquisition

The technical challenges students encounter during clinical training can significantly affect their skill acquisition, confidence and professional career. Studies indicate that when students are not exposed to fully operational medical devices or are relegated to peripheral roles, they graduate with gaps in competency (Bediako & Sarpong, 2020). These gaps translate into low confidence, especially when entering high-pressure clinical environments where fast, informed decisions about equipment are required.

According to Appiah-Kubi *et al*. (2023), more than 60% of final-year BME students in Ghana rated themselves as "partially prepared" or "unprepared" for immediate entry into professional roles due to limited hands-on training. The lack of regular feedback from supervisors during internships further inhibits self-assessment and correction, leading to a poor sense of technical identity (Ofori-Boateng & Anane, 2021).

Moreover, this lack of exposure and experience can impact students' enthusiasm and long-term commitment to the profession. Many become disillusioned about their career paths, especially if their expectations about contributing meaningfully to healthcare technology management are not met during training (Acheampong & Owusu, 2022).

2.8 Professional Development

Professional development in biomedical engineering is complex, comprising technical competency, ethical understanding, teamwork and communication skills. Clinical rotations are meant to foster these attributes by involving students in a dynamic healthcare environment. However, when students encounter technical barriers such as inaccessibility to tools, non-functional devices or a lack of real involvement in maintenance tasks their confidence and professional growth may be adversely affected (Mousavi *et al.,* 2019).

Evidence suggests that students who face such challenges during their clinical rotations often report lower levels of preparedness for industry practice upon graduation (Rivera *et al.*, 2021). In Ghana, where the biomedical engineering profession is still developing, these limitations can have long-term implications on the quality and capability of the workforce entering the healthcare system.

CHAPTER THREE

3.0. MATERIALS AND METHODS

3.1 Introduction

This chapter outlines the systematic procedures employed in conducting this research, ensuring reliability, validity and ethical rigor. It discusses the research design, data collection techniques, sampling strategies, study sites, target population, sample size, inclusion criteria, ethical considerations, data analysis and management and the instruments utilised.

3.2 Research Design

The study adopts a descriptive cross-sectional survey design using a structured questionnaire to collect quantitative and qualitative data from Biomedical Engineering students who have completed or are currently undertaking clinical rotations. A descriptive design was chosen because it allows for the systematic description of existing phenomena-in this case, the technical challenges encountered during clinical training-without manipulating the study environment. The cross-sectional approach provided a snapshot of the experiences of BME students across different universities in Ghana at a specific point in time.

The inclusion of both quantitative and qualitative data enhanced the richness of the findings. Quantitative data offered statistical insights into the availability and functionality of medical equipment and the frequency of challenges, while qualitative data provided deeper context regarding students' perceptions, confidence levels and readiness for professional practice. This methodological triangulation improves the validity of the study outcomes

3.3 Data Collection Techniques

Data for this study were collected using a structured, self-administered online questionnaire. The questionnaire was developed to address the study objectives, focusing on equipment availability and functionality, technical challenges and the impact on students' skills and readiness for professional practice. It was created using a digital platform (e.g., Google Forms), which allowed for easy access and submission. The link to the questionnaire was shared with participants via email, WhatsApp and university communication channels to reach a wide group of eligible Biomedical Engineering students across selected universities.

A snowball sampling technique was employed in the distribution of the questionnaire. Initially, the survey link was shared with a selected group of Biomedical Engineering students who had undergone clinical rotations. These initial participants were then requested to forward the link to other eligible peers within their academic networks. This approach helped to reach a broader population of students across various universities, particularly those who might not be easily accessible through direct channels.

All responses were submitted electronically and automatically recorded by the survey platform, ensuring confidentiality and efficient data management.

3.4 Sampling Technique

A purposive and stratified random sampling technique was adopted. First, purposive sampling was applied in selecting universities that offer Biomedical Engineering programs in Ghana. This ensured the study focused on institutions directly linked to the research objectives.

Second, within each selected university, stratified random sampling was used to select students. This technique ensures that only students with relevant exposure to the clinical training environment are included in the study, thereby enhancing the accuracy and relevance of the data collected. The stratification was based on academic year (e.g., second year, third year, and final year students), since students at different levels of training may encounter different types of challenges during their clinical rotations. Stratified random sampling was chosen to increase representativeness and reduce sampling bias.

3.5 Study Site

The study was carried out in universities in Ghana offering Biomedical Engineering programs, as these institutions serve as the foundation for students' academic and clinical training. The specific study sites included:

- Kwame Nkrumah University of Science and Technology (KNUST) – Kumasi.
- University of Ghana (UG) – Legon, Accra.
- Koforidua Technical University (KTU) – Koforidua.
- All Nations University (ANU) – Koforidua.

These institutions were selected because they provide accredited Biomedical Engineering programs and maintain partnerships with healthcare facilities for clinical placements. They represent a diversity of contexts public universities, private universities and technical university, thus offering a comprehensive picture of the training landscape in Ghana.

The inclusion of multiple institutions ensures variability in experiences, as resources, laboratory facilities and partnerships with hospitals differ across schools. This diversity strengthens the generalisability of the study findings to the broader population of Biomedical Engineering students in Ghana.

3.6 Target Population

The target population for this study consisted of all Biomedical Engineering students enrolled in Ghanaian universities offering accredited Biomedical Engineering programs. The study focused on students in their second, third and fourth (final) years, as these cohorts are directly involved in laboratory practice, industrial attachments and clinical training rotations. First-year students were excluded because they have minimal or no exposure to clinical training.

3.7 Sample Size

A total of 100–200 students were targeted to ensure a representative sample and reliable findings. The sample size was determined based on the estimated student population and expected response rate, allowing for meaningful statistical analysis.

The sample size was determined using Yamane's (1967) formula for finite populations:

- n=N/1+N(e^2)

Where:

- n = sample size
- N = total population
- e = margin of error (0.05)

Based on preliminary data obtained from institutional records, the combined population of Biomedical Engineering students across the four universities was approximately 171. Applying Yamane's formula:

- n=171/1+171(0.05^2)

- n=171/1+0.4275
- n=120

Thus, a minimum sample of 120 students was targeted.

3.8 Inclusion Criteria

The following inclusion criteria guided participation in the study:

- Participants must be officially enrolled in Biomedical Engineering programs in Ghana.
- Students must be in second year or higher, with some level of exposure to laboratory practice or clinical training.
- Willingness to provide informed consent before participation.

Exclusion criteria included first-year students and individuals not directly involved in Biomedical Engineering education.

3.9 Ethical Considerations

Ethical integrity was prioritised throughout the research process. The following ethical principles guided the study:

- Informed Consent: All participants were provided with detailed information about the purpose of the study, expected duration, potential risks and digital informed consent was obtained before they could proceed with the questionnaire.
- Confidentiality and Anonymity: Participants' responses were coded and no personal identifiers were included in the reporting of findings. Data were stored securely and used solely for academic purposes.
- Right to Withdraw: Participants had the right to withdraw from the study at any stage without penalty.
- Non-Maleficence: The study design ensured that no participant was exposed to harm or discomfort.

3.10 Data Analysis and Management

Collected data were coded and entered into Microsoft Excel, SPSS for analysis. Descriptive statistics such as frequencies, percentages, and mean scores were used to summarise the data.

Data Management: Soft copies were stored on password-protected devices. Only the research team had access to the data.

3.11 Research Instruments

A structured questionnaire was used as the primary data collection instrument. The questionnaire included both closed-ended (Likert scale and multiple-choice) and open-ended items to allow for both quantitative analysis and qualitative insights, items divided into four sections:

a. Demographic characteristics of the respondents
b. Medical equipment availability and functionality
c. Technical challenges encountered during clinical training
d. Professional development outcomes (skills, confidence, readiness).

The questionnaire was pre-tested for reliability and validity among a small group of students prior to full-scale deployment.

3.12 Methods

3.12.1 Demographic Distribution of Respondents

Respondents provided information on age, gender, level of study, institution and the duration and setting of their clinical rotation. This data helped frame the responses and allowed for subgroup analyses.

3.12.2 Equipment Availability and Functionality

Respondents were asked to rate the availability and working condition of medical equipment encountered during their clinical rotation. This included diagnostic (e.g. ECG), therapeutic equipment (e.g. infusion pump), patient monitoring devices imaging equipment (e.g. X-ray) and life support devices monitoring devices. Frequency of exposure and operational status (functional, non-functional, outdated) were recorded.

3.12.3 Technical Challenges Faced

Participants identified common technical issues encountered during training. These included limited access to equipment, insufficient training, difficulty operating medical equipment, inadequate supervision or guidance and limited time for practical experience. The impact of

these challenges on their ability to interact with and learn from the equipment were assessed and rated.

3.12.4 Professional Development

Students rated the extent to which the clinical experience contributed to their technical skills acquisition in troubleshooting, maintenance, calibration, confidence in handling biomedical equipment and overall readiness for professional practice. Specific questions examined whether technical difficulties impeded or enhanced their learning experience

CHAPTER FOUR

4.0 RESULTS AND DISCUSSION

4.1 Introduction

This chapter shows the results of the study on the problems biomedical engineering students face during their clinical training in Ghana and how these problems affect their learning and professional growth. The main aim of this chapter is to present and explain the information collected from the students so that the research questions and objectives can be answered.

The chapter is divided into sections based on the study objectives. These include the availability and working condition of medical equipment, the technical challenges students face while on clinical practice, and how these challenges affect their skills and professional development. The data is shown using charts, and simple explanations to make it easy to understand.

The findings in this chapter help to understand the difficulties students experience in clinical training and point out areas where improvements can be made to support their learning and future work as biomedical engineers.

4.2 Background Information of Respondent

4.2.1 Age of Respondent

The figure 1 shows the age distribution of the 120 respondents who participated in the study. From the data, it is evident that the majority of the students (66.67%) fall within the 20–24 years age group. This indicates that most biomedical engineering students in Ghana undertake clinical training in their early twenties, which aligns with typical tertiary education timelines.

The 25–29 years age group constitutes 20.83% of the respondents, while the under 20 years and 30 years and above groups make up 6.67% and 5.83% respectively. This suggests that only a small number of students are either younger than 20 or older than 30 when undergoing clinical training.

The mean age of all respondents is approximately 23.36 years, showing that the average student is in their early twenties. This reflects that most students gain practical clinical experience at a young adult stage, which is important for their professional development and learning capacity.

Overall, the age distribution indicates a concentrated student population in the 20–24 years category, which may have implications for tailoring clinical training programs to meet the needs of this age group. This is the graphical presentation of the data.

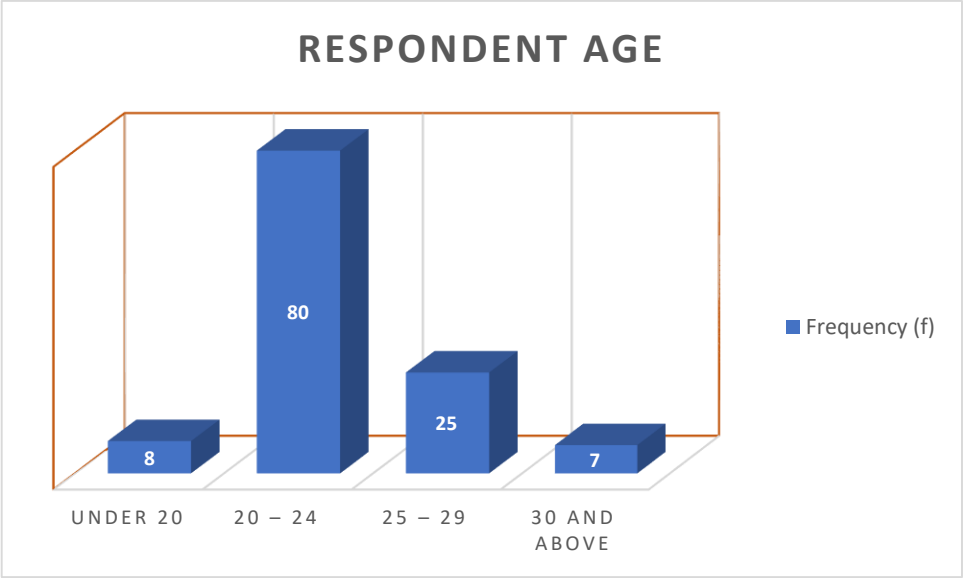

Figure 1

4.2.2 Gender of Respondents.

The figure 2 shows that out of 120 respondents, 66 (55%) were male and 54 (45%) were female. This indicates that there are slightly more male biomedical engineering students undergoing clinical training in Ghana than females.

The gender distribution is relatively balanced, with both sexes well represented, which suggests that the study's findings reflect the perspectives of both male and female students. Understanding gender distribution is important for analyzing whether technical challenges

during clinical training may affect male and female students differently. The figure below show the distribution of sex between the respondent

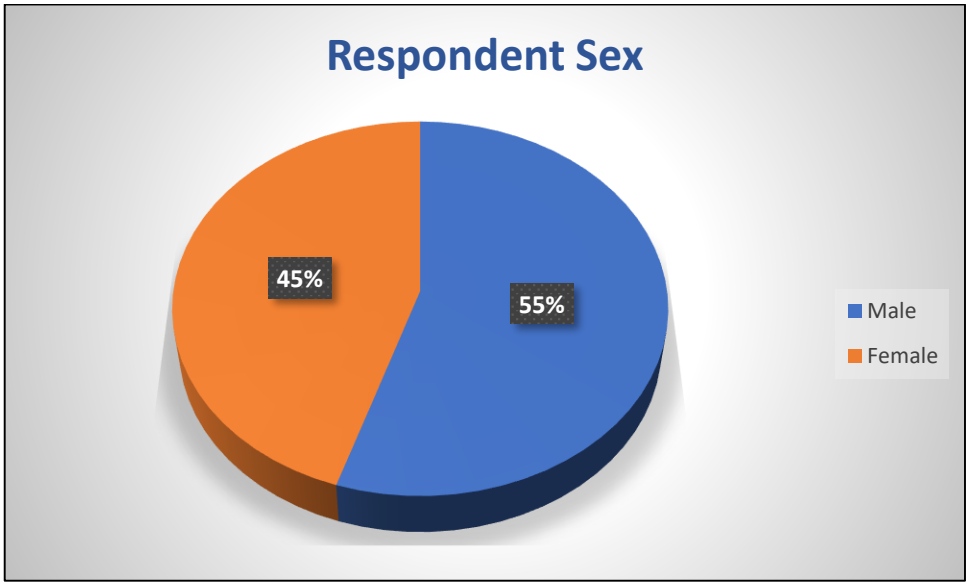

Figure 2

4.2.3 Educational Institution of the Respondent

It is observed that the majority of respondents (45.83%) are from Koforidua Technical University, making it the most represented institution in the study. All Nations University follows with 19.17%, while KNUST and University of Ghana account for 14.17% and 12.50% respectively. The remaining 8.33% are from other institutions.

This shows that the study drew participants from several universities across Ghana, but the findings may be influenced more by the experiences of students from KTU due to their higher representation. Below is the bar graph of the data.

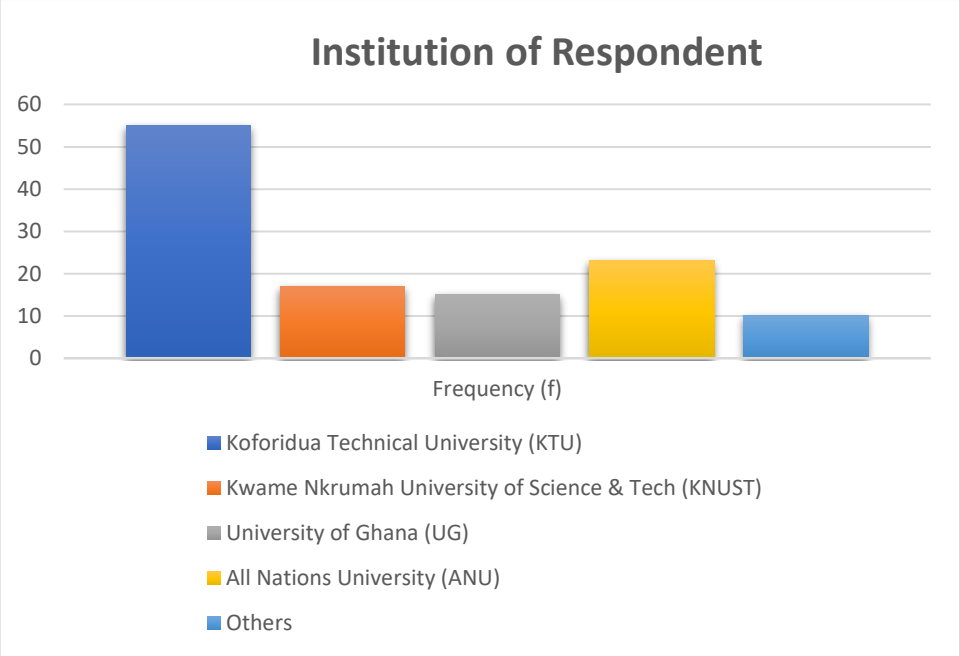

Figure 3

4.2.4 Respondents Level of Education

It is observed that the majority of respondents (54.17%) are in 3rd year, indicating that most biomedical engineering students undergoing clinical training are in their third year of study. The 2nd year students make up 30%, while 4th year students constitute 10.83%. Only a small proportion (5%) are in the 1st year.

This distribution suggests that clinical training is primarily undertaken by students in their middle years of study, likely when they have acquired sufficient theoretical knowledge to apply in practical settings. It also indicates that the study's findings will mostly reflect the experiences of 3rd-year students, with some input from other levels. The pie chart shows the distribution

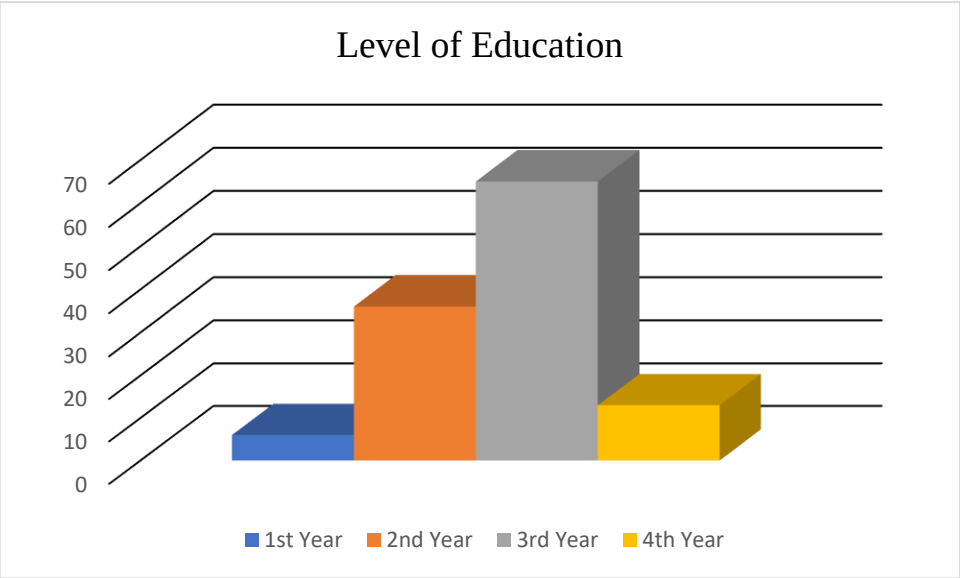

Figure 4

4.3. Objective 1: Assess the availability and functionality of medical equipment used during students' clinical training

This represent the availability and functionality of medical equipment during students clinical training.

Availability of the equipment

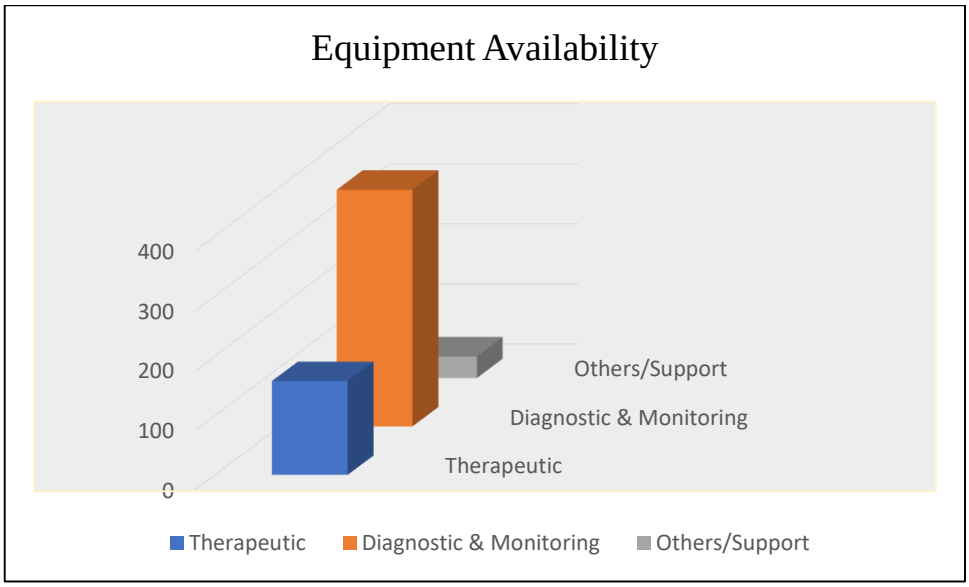

Figure 5

From figure 5, it shows that Diagnostic and Monitoring equipment were the most available during students' clinical training, with a total frequency of 395 (67.1%) out of 589. This means that students had more access to equipment like ECG machines, X-rays, ultrasound, pulse oximeters, and patient monitors. The high availability of these tools suggests that students had better opportunities to practice diagnostic and patient-monitoring activities.

Therapeutic equipment accounted for 158 (26.8%), showing that treatment-related devices such as infusion pumps, ventilators, defibrillators, and oxygen concentrators were present but less available compared to diagnostic and monitoring tools. This may have limited students' practical exposure to lifesaving procedures, especially in emergency and critical care situations.

Others/Support equipment such as autoclaves and centrifuges recorded the lowest availability with 36 (6.1%). These are essential in laboratory and sterilization processes, and their limited access may have affected students' understanding of infection control and laboratory-related practices.

Overall, the findings reveal that while diagnostic and monitoring devices were relatively available for students' use, therapeutic and support equipment were less accessible. This imbalance means that students may have gained more practice in patient observation and diagnosis but had fewer opportunities to build confidence in performing therapeutic interventions and laboratory support procedures.

This aligns with previous study by (Adusei *et al.,* 2021) who noted that in many developing countries, including Ghana, there is often limited access to modern, fully functional equipment in public hospitals due to procurement delays, funding constraints and poor maintenance culture

Functionality of the equipment

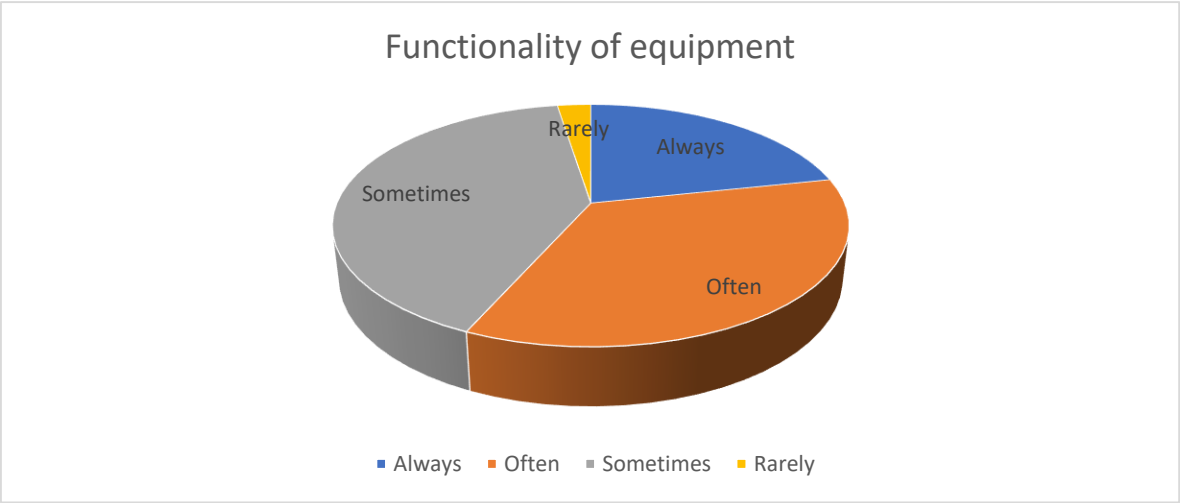

Figure 6

The figure 6 shows how frequently the medical equipment were functional during students' clinical training. The results reveal that 42.24% of the students indicated that equipment sometimes worked**,** while 36.21% said they often worked**.** Only 22.41% reported that the equipment were always functional**,** and a small proportion (2.59%) said the equipment rarely worked. These findings highlight that most of the equipment were not consistently functional throughout the training period.

The high percentages for "sometimes" and "often" suggest that students faced interruptions while using the equipment. Equipment may have broken down occasionally, required repairs, or were unavailable at certain times. Such irregular functionality likely limited students' ability to practice continuously and gain mastery over the machines. In situations where equipment was only partially functional, students might have had to observe rather than operate the machines themselves, reducing their hands-on experience and slowing down skill development.

Furthermore, the low proportion of students reporting that equipment was "always" functional indicates that fully reliable training tools were not consistently available. This could lead to reduced confidence among students, as they may feel unprepared when faced with real-life clinical situations. In addition, the lack of consistently functioning equipment

may affect the development of practical problem-solving skills, as students do not get the opportunity to operate devices under normal conditions.

Overall, the findings suggest that the functionality of medical equipment during clinical training was not fully reliable, which could negatively influence students' learning outcome.

This finding is consistent with Agyemang *et al*. (2019), who reported that biomedical engineering students in Ghana often face inadequate access to functioning medical equipment in teaching hospitals. Similarly, Mensah and Osei (2020) found that limited availability of laboratory and clinical tools hinders practical learning for health and engineering students. These studies support the present finding that equipment limitations affect student learning and skill development.

4.4 Objective 2: Identify the technical challenges biomedical engineering students encounter during clinical training in healthcare settings.

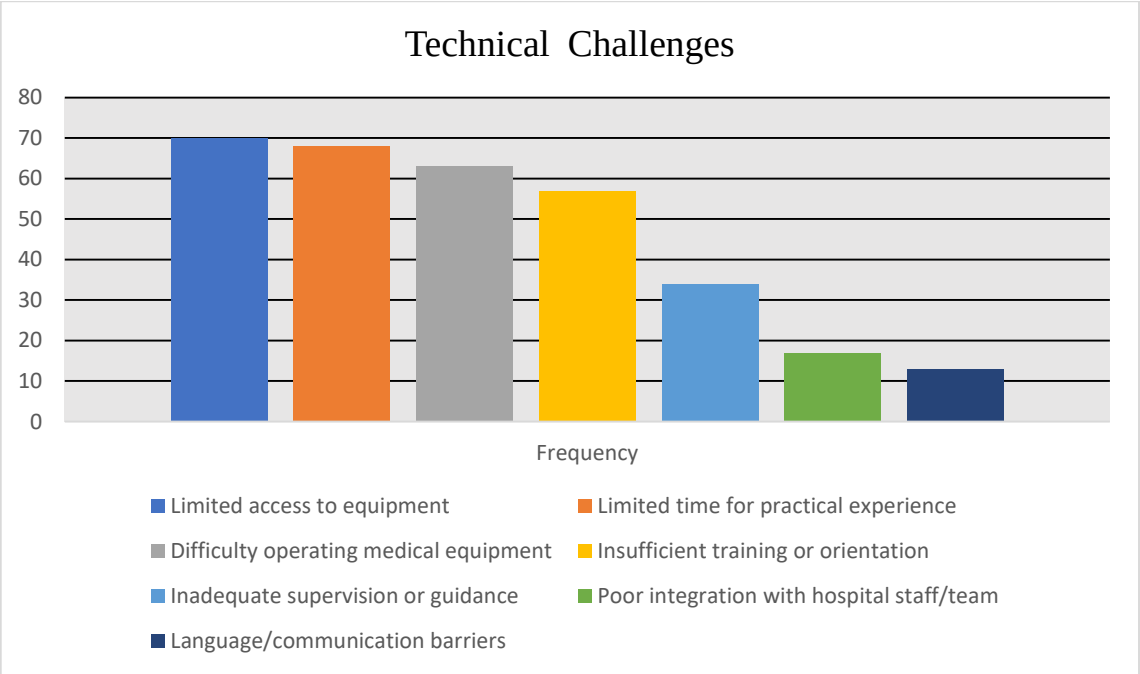

Figure 7

From figure 7, it shows that the most common technical challenge faced by students during clinical training was limited access to equipment (58.3%)**.** This means that many students could not regularly use the medical devices they needed to practice on. Without enough access, students may only observe equipment rather than actually operate it, which reduces the hands-on learning they need to gain confidence.

The second most common challenge was limited time for practical experience (56.7%)**.** This shows that even when equipment was available, students did not always get enough time to practice. Clinical training requires repeated practice to build skills, but short training schedules or too many students sharing the same equipment made it difficult for them to gain full experience.

Another key issue was difficulty operating medical equipment (52.5%)**.** Many modern medical devices have complex designs and interfaces. Without enough training and support, students often found it hard to operate them confidently. This also links to the challenge of insufficient training or orientation (47.5%)**,** where students felt that the instructions given before or during training were not enough to prepare them for real use of the machines.

Inadequate supervision or guidance (28.3%) was also reported as a challenge. This means that some students did not always have qualified staff or mentors available to guide them, leaving them uncertain about whether they were using the machines correctly.

Finally, fewer students reported challenges such as poor integration with hospital staff and teams (14.2%) and language or communication barriers (10.8%)**.** While these were less common, they still show that some students struggled to connect effectively with health professionals, which can make it harder to learn hospital procedures and teamwork.

Overall, these findings suggest that the biggest challenges to technical training were related to access, time, and training quality. Without enough equipment, enough time to practice, or strong guidance, students' ability to develop practical biomedical engineering skills was reduced.

Previous studies support these observations. Ofori and Appiah (2018) noted that biomedical engineering students often struggle with hands-on experience due to inadequate supervision and limited practical sessions. Boateng (2017) emphasized that technical challenges,

including unfamiliarity with advanced clinical devices, are common in engineering training programs in Ghana.

4.5 Objective 3: Analyze how these challenges affect students' skills acquisition, confidence and readiness for professional practice.

From the responses given by the students, it became clear that many challenges they faced during clinical training had a direct effect on their ability to gain practical skills, their confidence level, and their readiness for the health profession. The major challenges included difficulty in operating medical equipment, limited access to equipment, inadequate supervision, poor integration with hospital staff, insufficient training or orientation, limited time for practice, and communication barriers. All these challenges, when combined, had a strong effect on the way students experienced their clinical training.

4.5.1 Effect on Skills Acquisition

The first area affected is the acquisition of skills. Out of the 120 students, 41.7% reported that they were able to acquire skills and training, while the majority, 58.3%**,** said they did not. Students who reported limited access to equipment and limited time for practice were not able to practice as often as needed. For medical equipment such as ventilators, ECG machines, dialysis machines, and other complex devices, repeated practice is necessary to build competence. When access is restricted, students only observe rather than operate, which weakens their ability to develop hands-on skills. Some also mentioned difficulty in operating the equipment due to lack of proper guidance. Without close supervision and proper training, students cannot fully understand how to assemble, operate, or troubleshoot machines. This situation means that their learning process remains incomplete, and they graduate with gaps in their practical knowledge.

4.5.2 Effect on Confidence

The results show that 36.7% of students reported that they were not confident in their clinical practice, while 30.8% remained neutral. Only 9.2% felt confident, 20.8% were somewhat confident, and a very small group (2.5%) reported being very confident. This means that the majority of students (almost two-thirds) lacked strong confidence in their clinical abilities. Confidence in clinical practice usually comes from practical exposure, repeated experience, and guidance from supervisors. However, many students reported that they received

insufficient training and inadequate supervision, which made them unsure of their actions when operating medical equipment. Some also mentioned poor integration with hospital staff, which made them feel like outsiders instead of team members. These conditions create fear of making mistakes and weaken students' self-assurance. When students cannot freely practice and receive encouragement, they may begin to doubt their ability to perform well in real clinical situations. Since confidence is essential for quick decision-making and professional growth, its absence delays students' readiness to function effectively in hospital environments.

4.5.3 Effect on Readiness for Professional Practice

Finally, the results show that 36.9% of respondents said they were ready for the professional field, while 63.1% said they were not. This indicates that the majority of students do not feel fully prepared to transition from school into real hospital practice. Readiness means that a student can leave school and immediately apply what was learned in a clinical setting. When students face barriers such as lack of access to equipment, poor supervision, and insufficient hands-on practice, their readiness is delayed. They may graduate but still feel unprepared for the real demands of professional life. For instance, if a student has only seen a defibrillator or dialysis machine during attachment but never operated it, he or she will not feel prepared to use it confidently in emergencies. Similarly, without proper guidance, students may fail to learn standard safety practices, which are crucial in clinical environments. This lack of readiness affects not only the students but also the health system, since hospitals expect new graduates to contribute effectively.

This aligns with Nyarko (2019**)** and Asante (2016) who reported that biomedical engineering students' professional growth is affected when practical exposure is limited or when equipment is unreliable and that gaps in clinical training affect students' readiness to join the workforce, as they may lack critical problem-solving and hands-on skills respectively.

4.6 Summary

This chapter presented the analysis and interpretation of data collected from 120 biomedical engineering students in Ghana. The findings revealed important insights into the demographic characteristics of respondents and the challenges they face during clinical training.

From the demographic analysis, it was evident that most respondents were young adults aged 20–24 years, predominantly male, in their 3rd year of study, and largely enrolled at Koforidua Technical University. This context helped in understanding how the technical challenges impact students at different stages of their education.

Regarding the first objective**,** the study found that availability and functionality of medical equipment were significant challenges. Many students reported that some equipment was not always available or functional during clinical practice, which limited their practical learning opportunities.

In relation to the second objective, students faced technical challenges, including difficulty operating clinical devices, inadequate supervision, and limited hands-on experience. These challenges affected their ability to fully engage in clinical tasks and develop technical competence.

Finally, for the third objective, the study revealed that these technical challenges negatively affected professional development, reducing students' confidence, skill acquisition, and preparedness for future work in biomedical engineering.

Overall, the findings indicate that improvements in equipment availability, maintenance, and structured practical guidance are necessary to enhance the learning experience and professional growth of biomedical engineering students in Ghana. These conclusions provide a foundation for recommendations in the next chapter to address the identified challenges.

CHAPTER FIVE

5.0 CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter provides a summary of the major findings from the study, the conclusions drawn from those findings, and recommendations for improving the clinical training of biomedical engineering students in Ghana. The chapter brings together the results presented in Chapter Four and connects them to the study objectives. The main purpose is to highlight the key issues students face during clinical training, explain the implications of these issues on their professional development, and suggest measures to enhance the training process for future biomedical engineers.

5.2 Conclusion

The study concludes that biomedical engineering students in Ghana face serious challenges during clinical training that hinder their learning and professional development. While students gain sufficient exposure to basic and commonly available machines such as ECG and patient monitors, they have limited opportunities to work with advanced medical equipment. The inconsistent functionality of devices further reduces the quality of their training.

The technical challenges identified limited access to equipment, insufficient practice time, inadequate training and supervision, and difficulty in operating machines negatively affect students' ability to acquire skills, reduce their confidence levels, and delay their readiness for professional practice.

These findings show that clinical training environments in Ghana do not fully support the development of biomedical engineering students into competent professionals. Without addressing these gaps, graduates may continue to enter the workforce with incomplete skills, low confidence, and limited ability to handle the diverse challenges of modern healthcare systems.

5.3 Recommendations

To improve the clinical training of biomedical engineering students in Ghana, the following recommendations are proposed:

1. Universities

Universities need to go beyond the theory and create more opportunities for students to gain practical experience before they step into the hospital environment. This means investing in well-equipped laboratories where students can practice with real or simulated biomedical devices. Universities should also strengthen their partnerships with hospitals so that clinical rotations are well-structured, supervised, and purposeful. By regularly updating their training content to reflect modern technologies, universities can better prepare students to meet the realities of the healthcare system.

2. Hospitals

Hospitals play a key role in shaping the practical skills of students, and so they need to ensure that equipment is available and functional for training purposes. Assigning qualified biomedical engineers as mentors during clinical rotations would give students the guidance they need to troubleshoot, maintain, and calibrate devices with confidence. Hospitals should also prioritise preventive maintenance so that students are exposed to working equipment rather than broken-down devices. With proper supervision and structured schedules, hospitals can transform clinical rotations into a more meaningful and confidence-building experience for students.

3. Government

The government must recognise that biomedical engineers are essential to strengthening the healthcare system. Adequate funding should be provided to universities to improve training facilities and to hospitals to ensure that biomedical equipment is regularly upgraded and maintained. Policies that encourage collaboration between universities, hospitals, and industry will also help close the training gaps identified in this study. Finally, by supporting continuous professional development and certification programs, the government can ensure that Ghana produces biomedical engineers who are fully equipped to meet the country's growing healthcare technology needs.

5.4 Limitations of the Study

- Scope of respondents: The study was limited to 120 students from selected universities, meaning the findings may not fully represent all biomedical engineering students in Ghana.
- Self-reported data: Information was based on students' experiences and perceptions, which may include personal biases or memory inaccuracies.
- Focus on student perspectives only: The study did not capture the views of hospital staff, supervisors, or policymakers, who could provide additional insights into the challenges.
- Limited equipment coverage: While several devices were included, the study could not cover every type of medical equipment used in clinical training.

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7.0 APPENDIX

Table 4.0 *Respondent Age*

Age of Respondent			
Age Group (Years)	Frequency (f)	Percentage (%)	Mean of Age Group
Under 20	8	6.67	18
20 – 24	80	66.67	22
25 – 29	25	20.83	27
30 and above	7	5.83	32
Total / Overall	120	100	23.36

Table 4.1 *Sex of Respondent*

Respondents Age		
Gender	Frequency (f)	Percentage (%)
Male	66	55
Female	54	45
Total	120	100

Table 4.2 *Educational Institution of the Respondent*

School / Institution	Frequency (f)	Percentage (%)
Koforidua Technical University (KTU)	55	45.83
Kwame Nkrumah University of Science & Tech (KNUST)	17	14.17
University of Ghana (UG)	15	12.50
All Nations University (ANU)	23	19.17
Others	10	8.33
Total	120	100

Table 4.3 *Respondents Level of Education*

Education Level of Respondent		
Year / Level	**Frequency (f)**	**Percentage (%)**
1st Year	6	5.0
2nd Year	36	30.0
3rd Year	65	54.17
4th Year	13	10.83
Total	120	100

Table 4.4 *Availability of Medical Equipment*

Availability of Medical Equipment during Students' Clinical Training		
Equipment	Frequency	Percentage (%)
Electrocardiography	83	22.74
Infusion pumps	50	13.69
Pulse Oximeter	72	19.74
Defibrillator	20	11.52
Ventilator	10	11.25
Ultrasound Machine	64	17.54
X-ray	79	21.65
MRI	2	6.85
Dialysis machine	4	10.15
Patient Monitor Device	42	2.19
Suction machine	17	4.67
Autoclave	11	3.01
Centrifuge	25	1.10
Electrophoresis machine	3	0.82
Oxygen concentrator	41	2.74
BP apparatus	37	2.19

Electroencephalogram	8	5.49
Incubator	3	0.82
Infrared lamp	8	0.55
Anesthesia machine	5	1.37
Hematology analyzer	3	0.82
Chemistry analyzer	2	0.55

Table 4.5 *Functionality of Medical Equipment*

Functionality of Medical Equipment during Clinical Training		
Functionality	Frequency	Percentage (%)
Always	26	22.41
Often	42	36.21
Sometimes	49	42.24
Rarely	3	2.59
Total	120	100

Table 4.6 *Technical Challenges*

Technical Challenge	Frequency	Percentage (%)
Limited access to equipment	70	58.3%
Limited time for practical experience	68	56.7%
Difficulty operating medical equipment	63	52.5%
Insufficient training or orientation	57	47.5%
Inadequate supervision or guidance	34	28.3%
Poor integration with hospital staff/team	17	14.2%
Language/communication barriers	13	10.8%

Table 4.7

Category	Equipment	Frequency	Total
Therapeutic	Infusion pumps (50), Defibrillator (20), Ventilator (10), Dialysis machine (4), Suction machine (17), Oxygen concentrator (41), Anesthesia machine (5), Incubator (3), Infrared lamp (8)	158	
Diagnostic & Monitoring	Electrocardiography – ECG (83), Ultrasound machine (64), X-ray (79), MRI (2), Electrophoresis machine (3), Electroencephalogram – EEG (8), Hematology analyzer (3), Chemistry analyzer (2), Pulse oximeter (72), Patient monitor device (42), BP apparatus (37)	395	
Others/Support	Autoclave (11), Centrifuge (25)	36	
Grand Total			589